Sangjun Ji

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Research Interests

- Graph-based Machine Learning & Representation Learning
- Knowledge Graphs (Hyper-relational KG, KG Reasoning)
- Synergizing Graphs and LLMs (Recommendation, KGQA, Anomaly Detection)

Education

Konkuk University, Seoul, Republic of Korea *Mar. 2020 - Present*
B.S. in Computer Science and Engineering, (Expected graduation: Feb. 2027)
GPA: 4.36 / 4.5, Major GPA: 4.47 / 4.5

Experience

Data Mining Lab, KAIST AI *Jun. 2026 - Present*
Undergraduate Research Intern, (Advisor: Prof. Kijung Shin)

Graph & Language Intelligence Lab, Konkuk University *Jul. 2024 - Jun. 2026*
Undergraduate Research Intern, (Advisor: Prof. Byungkook Oh)

Team Lett, Pangyo, Republic of Korea *Oct. 2023 - Jul. 2024*
Own Business, Co-Founder at Lett, challenging tech entrepreneurship across apps and web services

Publications

[1] Internalizing Negation-Gated Logical Rules into LLMs for Document-Level Relation Extraction
- Hye-Yoon Baek, **Sangjun Ji**, Jimyeung Seo, Hae-Yoon Koo, Xiongnan Jin and Byungkook Oh*
- *Under Review*

[2] Joint Global-Local Representations via Relation-Entity Pair Encoding for Hyper-Relational Knowledge Graphs
- **Sangjun Ji**, Sangjune Kim, Bonyou Koo, Youngho Lee, Xiongnan Jin, and Byungkook Oh*
- *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2026*
- *Research Track (Acceptance Rate: 18.5%)*

[3] Mention-Context Hypergraph Aggregation for Document-level Relation Extraction
- **Sangjun Ji**, Hye-Yoon Baek, Donghyun Lee and Byungkook Oh*
- *Korea Computer Congress (KCC), 2026*
- *Undergraduate Paper Track*

Patents

[1] Method and Apparatus for Link Prediction of Hyper-Relational Facts on Hyper-Relational Knowledge Graphs
하이퍼관계형 지식 그래프 상의 하이퍼관계형 사실에 대한 링크 예측 방법 및 장치
- **Sangjun Ji** and Byungkook Oh
- *KR Patent, filed Dec. 12, 2025*

Awards

Kakao × KIISE AI Agent Competition (KSC 2025) *Dec. 2025*
- 카카오x한국정보과학회 AI Agent 경진대회
- Multi-Agentic Graph RAG for Fair Meet-up Location Recommendation and Schedule Coordination
- *Excellence Award (우수상, 3rd tier)*

Projects

KU래쪄용: AI-Powered Conflict Mediation Chat App *Konkuk Univ.*
KU래쪄용: 갈등해소 AI 채팅 앱 *Mar. 2026 - Present*
- Multi-Agent LLM 토론 시스템 기반 다툼 판결 기능 설계 · 구현
- LLM 기반 실시간 언어순화 기능 (공격적 표현 감지 · 완화)으로 갈등 격화 방지
- 대화 내 감정 신호를 정량화하는 감정온도계 기능으로 사용자 자기인식 지원

Graph-based Rule-aware Compliance Framework *Konkuk Univ.*
LLM 규칙 준수 일관성 향상을 위한 Graph-based Rule-aware Compliance Framework *Mar. 2026 - Jun. 2026*
- 정책 문서로부터 atomic rule을 추출하는 schema · extraction pipeline 설계 (DocRE · LLM 기반 방법 검토)
- Neo4j 기반 Rule Graph 구조 구현 (base · Semi-expanded · Full-expanded 단계적 확장)
- 외부 규칙 저장소를 LLM과 연계하여 규칙을 일관되게 준수하는 프레임워크 설계/구현

Multi-Agentic Graph RAG for Fair Meet-up Location Recommendation and Schedule Coordination *GLI Lab.*
공평한 만남을 위한 약속 장소 추천 및 일정 조율 Multi-Agentic GraphRAG *Oct. 2025 - Dec. 2025*
- Multi-Agentic GraphRAG 기반 약속 장소 추천 · 일정 조율 workflow 설계 · 구현
- Shared Memory graph DB로 user 선호/이력/문맥을 관리해 개인화된 추천 · 조율 지원
- 이동 시간 기반 공평성 및 예산/선호를 반영하고, PlayMCP로 톡캘린더 일정 생성까지 end-to-end 연동

GraphRAG with User-Item Interactions for Recommendation *Konkuk Univ.*
사용자–아이템 상호작용 기반 GraphRAG 추천 *Jul. 2024 - Aug. 2024*
- tutorial framework 구현 및 대체 dataset 지원을 위한 확장 개발

Pzzk: A Clean, Serverless iOS Calendar *Lett.*
Pzzk: 깔끔한 iOS 캘린더 앱 *Jan. 2024 - Feb. 2024*
- “the simplest calendar in the world” 콘셉트의 iOS 캘린더 앱 PzzK 개발 및 공식 출시
- server 없이 iCloud 동기화 기능 구현 (serverless iCloud sync)

Licenses & Certifications

- Craftsman Fork Lift Truck Operator (지게차운전기능사), HRD Korea (Q-Net) *Dec. 2022*